

In Figure 15.2, a dendrogram is constructed using SPSS which clusters sample data. A dendrogram is a branching diagram that represents the relationships of similarity among a group of entities. Each branch is called a clade. The terminal end of each clade is called a leaf. Clades can have just one leaf (these are called simplicifolious, a term from botany that means “single leafed”) or they can have more than one. Two-leaved clades are bifolious, three-leaved are trifolious, and so on. There is no limit to the number of leaves in a clade. The arrangement of the clades tells us which leaves are most similar to each other. The width of the branch points indicates how similar or different they are from each other: the greater the width, the greater the difference. In this particular dendrogram, we see that chunk 6, 10 and 13 are completely separate from all the others as they are clustered through a separate leaf of the clade. We interpret its placement as indicating that the distribution in that chunk is substantially different from the distribution in the remaining chunks. By closer inspection, it can be observed that these 3 individuals are post-graduates as well as employed on either contractual basis or unemployed i.e. it reflects that higher educational attainment has not improved their job status as well as job security.

If we are trying to identify which individual segments are most similar to each other, we read the dendrogram from the left to right, identifying the first clades to join together as we move from left to right. By closer inspection we find 4 clusters which have most similar objects. The first cluster is of 2 and 15 indicating better job status after graduation, the second cluster is of 1 and 3 (both are employers and post-graduates), the 3rd cluster is of 4,8,9 and 12 which has linked post-graduates and the regularity of their employment while the 4th cluster is made up of 6 and 10 indicating contractual work despite being post-graduate. The further the clades away from the observations/objects, less is the similarity between them.

Table 15.1: Similarity between level of education and level of employment

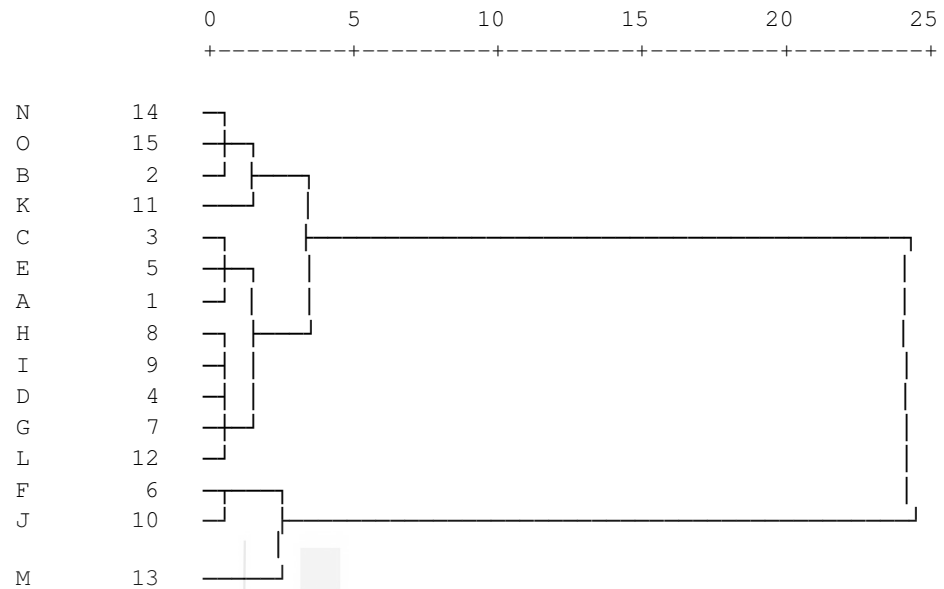
Proximity Matrix

Observation	Squared Euclidean Distance														
	1:A	2:B	3:C	4:D	5:E	6:F	7:G	8:H	9:I	10:J	11:K	12:L	13:M	14:N	15:O
1:A	.000	1.000	.000	1.000	.000	16.000	1.000	1.000	1.000	16.000	5.000	2.000	29.000	2.000	2.000
2:B	1.000	.000	1.000	2.000	1.000	17.000	2.000	2.000	2.000	17.000	4.000	5.000	26.000	1.000	1.000
3:C	.000	1.000	.000	1.000	.000	16.000	1.000	1.000	1.000	16.000	5.000	2.000	29.000	2.000	2.000
4:D	1.000	2.000	1.000	.000	1.000	9.000	.000	.000	.000	9.000	2.000	1.000	20.000	1.000	1.000
5:E	.000	1.000	.000	1.000	.000	16.000	1.000	1.000	1.000	16.000	5.000	2.000	29.000	2.000	2.000
6:F	16.000	17.000	16.000	9.000	16.000	.000	9.000	9.000	9.000	.000	5.000	10.000	5.000	10.000	10.000
7:G	1.000	2.000	1.000	.000	1.000	9.000	.000	.000	.000	9.000	2.000	1.000	20.000	1.000	1.000
8:H	1.000	2.000	1.000	.000	1.000	9.000	.000	.000	.000	9.000	2.000	1.000	20.000	1.000	1.000
9:I	1.000	2.000	1.000	.000	1.000	9.000	.000	.000	.000	9.000	2.000	1.000	20.000	1.000	1.000
10:J	16.000	17.000	16.000	9.000	16.000	.000	9.000	9.000	9.000	.000	5.000	10.000	5.000	10.000	10.000
11:K	5.000	4.000	5.000	2.000	5.000	5.000	2.000	2.000	2.000	5.000	.000	5.000	10.000	1.000	1.000
12:L	2.000	5.000	2.000	1.000	2.000	10.000	1.000	1.000	1.000	10.000	5.000	.000	25.000	4.000	4.000
13:M	29.000	26.000	29.000	20.000	29.000	5.000	20.000	20.000	20.000	5.000	10.000	25.000	.000	17.000	17.000
14:N	2.000	1.000	2.000	1.000	2.000	10.000	1.000	1.000	1.000	10.000	1.000	4.000	17.000	.000	.000
15:O	2.000	1.000	2.000	1.000	2.000	10.000	1.000	1.000	1.000	10.000	1.000	4.000	17.000	.000	.000

Source: Assessment of Human Wellbeing: A Multidimensional Approach, a research study (unpublished) undertaken in 2015 by Prof. Narayan Prasad.

DENDROGRAM USING WARD METHOD

Rescaled Distance Cluster Combine

**Fig.15.2: Dendrogram**

15.7 OTHER APPROACHES: TWO-STEP CLUSTER ANALYSIS

Among the other widely used non-hierarchical approaches is the approach of Two-step cluster analysis. The Two Step Cluster Analysis procedure is an exploratory tool designed to reveal natural groupings (or clusters) within a data set that would otherwise difficult to find. The Algorithm employed by this procedure has several desirable features that differentiate it from traditional clustering techniques:

- The ability to create clusters based on both categorical and continuous variables.
- Automatic selection of the number of clusters.
- The ability to analyze large data files efficiently.

In order to handle categorical and continuous variables, the TwoStep Cluster Analysis procedure uses a likelihood distance measure which assumes that variables in the cluster model are independent. Further, each continuous variable is assumed to have a normal (Gaussian) distribution and each categorical variable is assumed to have a multinomial distribution.

The two steps of the Two Step Cluster Analysis procedure's algorithm can be summarized as follows:

Step 1 The procedure begins with the construction of a Cluster Features (CF) Tree. The tree begins by placing the first case at the root of the tree in a leaf node that contains variable information about that case. Each successive case is then added to an existing node or forms a new node, based upon its similarity to existing nodes and using the distance measure as the similarity criterion. A

node that contains multiple cases contains a summary of variable information about those cases. Thus, the CF tree provides a capsule summary of the data file.

Step 2 The leaf nodes of the CF tree are then grouped using an agglomerative clustering algorithm. The agglomerative clustering can be used to produce a range of solutions. To determine which number of clusters is “best”, each of these cluster solutions is compared using Schwarz's Bayesian Criterion (BIC) or the Akaike Information Criterion (AIC) as the clustering criterion.

15.8 INTERPRETATION OF THE RESULTS

We shall now discuss this method with the help of an example based on 416 observations collected through primary survey for the purpose of research study entitled “Assessment of Human Wellbeing in Delhi :Multidimensional Approach”. The researcher in this study attempted to find out the clusters on the basis of household level of income and their educational status. Four income levels have been used to classify the sample into four categories: income up to 2.5 lakhs, between 2.5 and 5 lakhs, between 5-10 lakhs and above 10 lakhs. The educational attainment has been taken as : illiterate (no education), upto middle education level, between middle to higher secondary education, upto graduation, upto post-graduation and PhD and above. The hypothesis was that higher the level of education, higher will the level of income meaning thereby a positive association between level of education and the level of income. Let us examine this hypothesis by applying the two-step cluster analysis technique.

Figure 15.3 shows results of cluster 1 wherein educational attainment of post-graduation has been found to be inter-linked with income category of 10 lakhs and above. This is logically also true as higher levels of educational attainment are associated with higher income levels. In **Figure 15.4**, it can be observed that graduation is associated with income-levels of 5-10 lakhs and these two sub-groups have been clustered together.

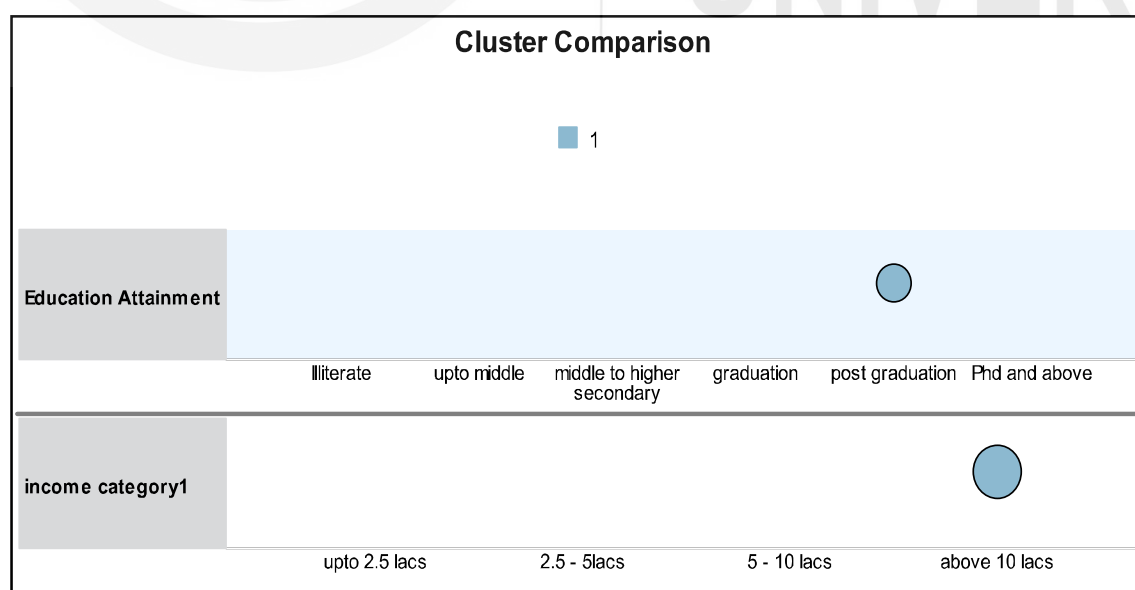


Fig.15. 3: Cluster 1

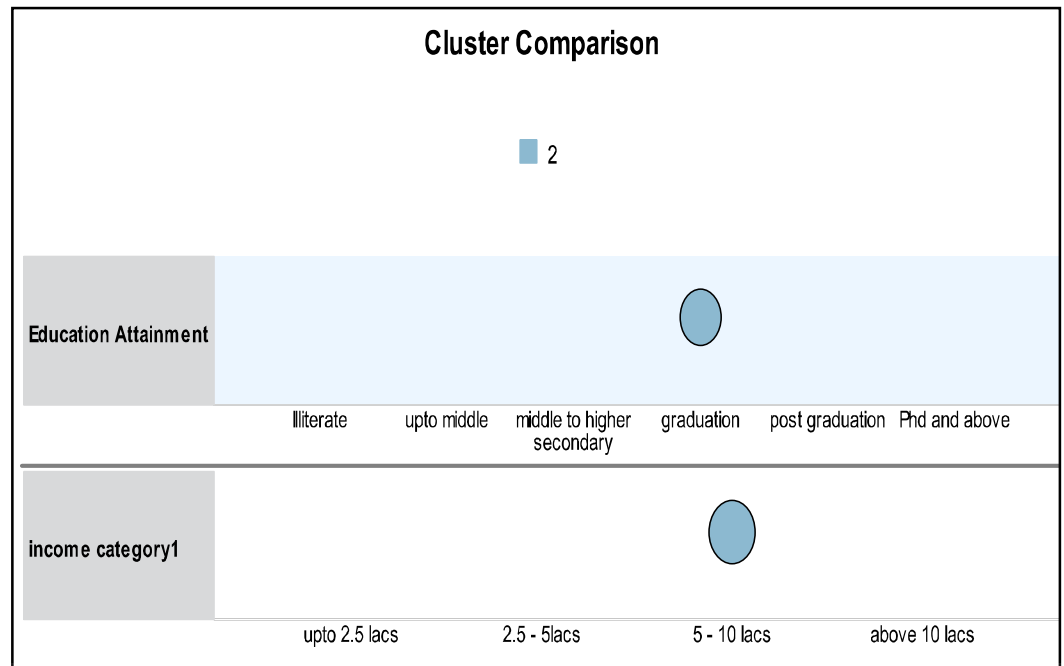


Fig.15.4: Cluster 2

Figure 15.5 finds another cluster linking illiterate individuals with the lowest income category (i.e. less than 2.5 lakhs) while **Figure 15.6** finds the fourth cluster consisting of middle to higher secondary education and income level upto 2.5 lakhs. Thus intuitively also, two-step cluster analysis has formed clusters of similar attributes.

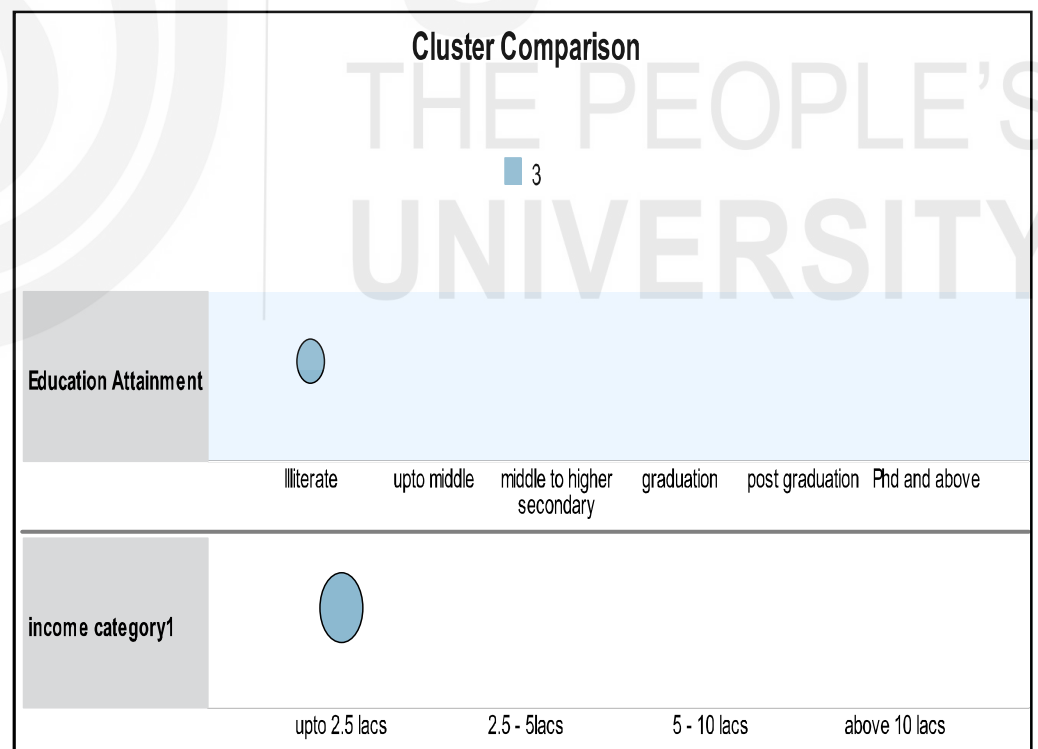


Fig.15.5: Cluster 3

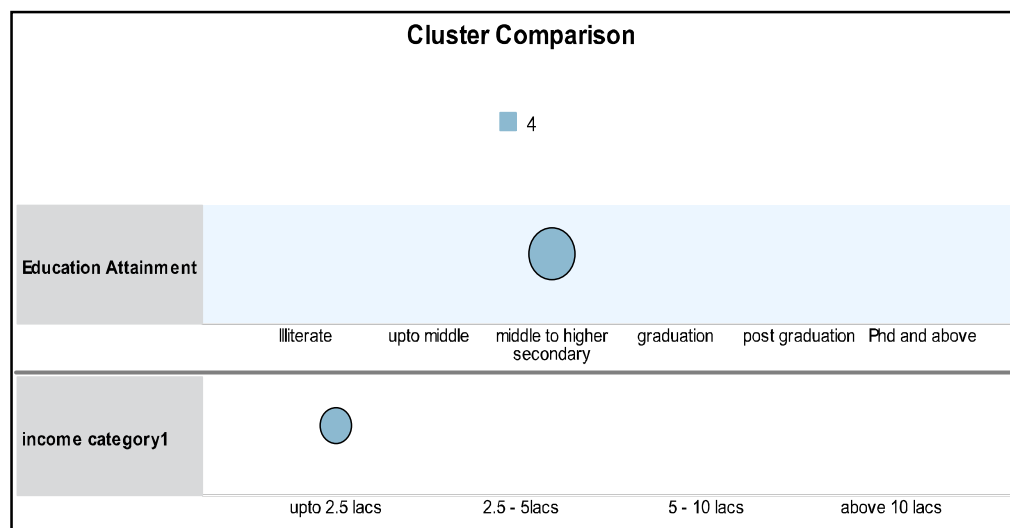


Fig.15.6: Cluster 4

Thus, by using Two step cluster analysis technique , we have separated 416 individuals into 4 broad clusters based on their income category and level of educational attainment which, in turn , has enabled us to test the hypothesis whether an association exist between the level of education and level of income. Thus the hypothesis that higher the level of education, higher the level of income has been accepted by the results of the cluster analysis.

Check Your Progress 2

- 1) What are the hierarchical methods used in cluster analysis? Critically explain how dendrograms help in visual interpretation of clustered data.

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- 2) Explain the application of two-step cluster analysis in research. What are its advantages?

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15.9 LET US SUM UP

The objective of cluster analysis is to assign observations to groups (clusters) so that observations within each group are similar to one another with respect to variables or attributes of interest and the groups themselves stand apart from

one another. Cluster analysis embraces a variety of methods aiming to group observations or variables into homogeneous and distinct clusters. For groupings based on variables, frequently used measures of the similarity of observations are the Euclidean. Squared Euclidean are applied to the original, standardized, or weighted variables. For groupings based on attributes, a measure of the similarity of two observations is the ratio of the number of matches (identical categories) to the number of attributes.

Hierarchical methods begin with as many clusters as there are observations and end with a single cluster containing all observations. All clusters formed by these methods are mergers of previously formed clusters. Other types of clustering methods are the hierarchical divisive (beginning with a single cluster and ending with as many clusters as there are observations) and the non-hierarchical methods. In recent years, some other clustering methods have also evolved like the two-step cluster analysis discussed above which are efficient in handling large data sets of continuous and categorical variables as well. Cluster analysis technique gives a different angle of viewing and interpreting data which may help the researcher to discover significant new relationships between the variables used in a study.

15.10 KEY WORDS

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|-----------------------------|---|--|
| Cluster analysis | : | It is the task of grouping a set of objects in such a way that objects in the same group (called a cluster) are more similar (in some sense or another) to each other than to those in other groups (clusters). |
| Agglomerative method | : | In which subjects start in their own separate cluster. The two 'closest' (most similar) clusters are then combined and this is done repeatedly until all subjects are in one cluster. At the end, the optimum number of clusters is then chosen out of all cluster solutions. |
| Divisive method | : | In which all subjects start in the same cluster and the above strategy is applied in reverse until every subject is in a separate cluster. Agglomerative methods are used more often than divisive methods, so this handout will concentrate on the former rather than the latter. |
| Dendrogram | : | When carrying out a hierarchical cluster analysis, the process can be represented on a diagram known as a dendrogram. This diagram illustrates which clusters have been joined at each stage of the analysis and the distance between clusters at the time of joining. |
| Centroid method | : | Here the centroid (mean value for each variable) of each cluster is calculated and the distance between centroids is used. Clusters whose centroids are closest together are merged. This method is also fairly robust. |

: In this method all possible pairs of clusters are combined and the sum of the squared distances within each cluster is calculated. This is then summed over all clusters. The combination that gives the lowest sum of squares is chosen. This method tends to produce clusters of approximately equal size, which is not always desirable. It is also quite sensitive to outliers. Despite this, it is one of the most popular methods, along with the average linkage method.

K-Means Clustering

: The *k*-means algorithm was popularized and refined by Hartigan (1975). The basic operation of that algorithm is relatively simple. Given a fixed number of (desired or hypothesized) *k* clusters, assign observations to those clusters so that the means across clusters (for all variables) are as different from each other as possible.

Euclidean Distance

: The straight line distance between two points in a Cartesian coordinate system. The Euclidean distance can be determined using the Pythagorean Theorem. In two dimensions, the Euclidean distance is $[(x_1 - x_2)^2 + (y_1 - y_2)^2]^{0.5}$. Usually, the points represent samples and the axes of the Cartesian coordinate system represent the abundances of species.

Exploratory Analysis

: A general term for an analysis in which the chief objectives is to find pattern in the data. Often, exploratory analysis conflicts with hypothesis testing. For example, stepwise regression is permissible in exploratory analysis, but can cause serious problems if you are interested in testing hypotheses.

15.11 SOME USEFUL BOOKS

- 1) Data Mining, Concepts and Techniques: Jiawei Han, Micheline Kamber, Jian Pei, 3rd Edition, Morgan Kaufmann, 2012.
- 2) Cluster Analysis of Economic Data, Hana Rezankova, Statistika, 94 (1), Czech Republic, 2014
- 3) <http://wheatoncollege.edu/lexomics/files/2012/08/How-to-Read-a-Dendrogram-Web-Ready.pdf>
- 4) https://en.wikipedia.org/wiki/Cluster_Analysis
- 5) <https://arifkamarbafadal.files.wordpress.com/2011/09/ebook-038-tutorial-spss-two-step-cluster-analysis.pdf>

15.12 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) See Section 15.2 and 15.3
- 2) See Section 15.4 and 15.5

Check Your Progress 2

- 1) See Section 15.6
- 2) See Section 15.7

15.13 EXERCISES

- 1) For what purposes and situations would you choose to undertake cluster analysis?
- 2) Explain in your own words briefly what hierarchical cluster analysis does. Check your answer against the material above.
- 3) Explain the difference between hierarchical methods and non-hierarchical methods of clustering citing their advantages and disadvantages.
- 4) Is clustering same as classification? What is the essential difference between classification and clustering?

UNIT 16 CORRESPONDENCE ANALYSIS

Structure

- 16.0 Objectives
- 16.1 Introduction
- 16.2 Correspondence Analysis: Concept and Its Features
- 16.3 Steps and Algorithm Involved in Correspondence Analysis Technique
- 16.4 Basic Concepts and Definitions
 - 16.4.1 Primitive Matrix
 - 16.4.2 Profiles
 - 16.4.3 Masses
 - 16.4.4 Correspondence Matrix
 - 16.4.5 Augmented Correspondence Matrix
 - 16.4.6 Inertia
 - 16.4.7 Distances
- 16.5 Reduction of Dimensionality
- 16.6 Biplots
- 16.7 Interpretation of the Results of Correspondence Analysis
- 16.8 Multiple Correspondence Analysis
- 16.9 Let Us Sum Up
- 16.10 Key Words
- 16.11 Some Useful Books
- 16.12 Answers or Hints to Check Your Progress
- 16.13 Exercises

16.0 OBJECTIVES

After going through this unit, you will be able to:

- state the concept of Correspondence Analysis (CA) as a special case of principal component analysis;
- discuss the special features and applications of CA technique;
- explain the computational algorithm of Correspondence Analysis;
- elucidate the process of interpretation of Correspondence Analysis results; and
- learn how to apply Correspondence Analysis technique in conducting research in social sciences.

16.1 INTRODUCTION

Understanding data is the ultimate problem of statistics and no one technique/method can satisfactorily answer all our curiosities regarding any particular set of data. Given a set of data, we often formulate hypothesis regarding it. This formulation is motivated by our information regarding the variables of the data. We then apply various statistical tools to test our hypothesis.

In such traditional methods of hypothesis testing, we verify a priori hypothesis, that is, the hypothesis pre-assumed regarding the relations between variables involved in the data. However, these methods naturally suffer from a limitation. The pre-assumed hypotheses may not be the only possibility. There can be other hidden relationships among the variables of the data. How do we find them?

You have studied in Unit 1 and 8 that from measurement scale point of view, data can be of four types – nominal, ordinal, interval and ratio scale. Principal Component Analysis (PCA) technique is used to identify the relative importance of factors or components responsible for variance in the dependent variable. PCA, however, pre-supposes tables consisting of continuous measurement (i.e. interval or ordinal scale of data). If a situation arises when one is interested to know association between two or more categorized variables, search for an alternative analytic technique is needed. For example, you may be interested to examine how the level of education is related to the nature of employment (i.e. regular, casual, contract etc.). Similarly how the quality of employment is associated with different levels of income? In such situation, you need a data analytic technique which can help you in detecting structural relationships among the categorized variables. Correspondence Analysis (CA) is analytic technique which helps to examine such structural relationship. This technique is very useful to analyze simple two way and multi-way tables containing some measure of correspondence between the rows and columns. Hence, in this unit, we shall throw light on various concepts used in correspondence analysis, its pre-requisites, its computation algorithm, interpretation of its results, etc. Let us begin with stating the meaning of correspondence analysis.

16.2 CORRESPONDENCE ANALYSIS: CONCEPT AND ITS FEATURES

Correspondence Analysis (CA) is a descriptive exploratory technique of multivariate statistical analysis, allowing to define the nature and structure of the relationship between qualitative variables measured in nominal and ordinal scales. CA is a way of seeing the association in a two way – cross tabulation rather than measuring it. In general, CA as a method is used to decomposes the overall chi-square statistics by defining a system with a smaller number of dimensions, in which the deviation from the expected values are presented. In brief, CA may be defined as a special case of Principal Component Analysis of the rows and columns of a table especially applicable to a cross tabulation. The primary goal of CA is to transform a table of numerical information into a graphical display, in which each row and each column is depicted as a point.

The important features of correspondence analysis are:

- 1) CA provides the multivariate treatment of the data through simultaneous consideration of multiple categorical variables.
- 2) This technique can reveal relationships that would not be detected in a series of pair wise comparison of variable.
- 3) It also provides graphical display of row and column points in biplots to detect structural relationships among the variable categories and objects.